Front-End Implementation Using Jetpack Compose

Frontend Compose Report

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1 Introduction

This report documents the front-end implementation work completed using Jetpack Compose in Android Studio. The purpose of this work was to understand how modern Android user interfaces are built using composable functions and layout components.

Jetpack Compose is a declarative UI toolkit that allows developers to create Android user interfaces using Kotlin instead of XML layouts. It simplifies UI development by allowing developers to build reusable UI components and combine them to create application screens.

2 Understanding Composable Functions

Composable functions are the fundamental building blocks used to define user interfaces in Jetpack Compose. A composable function describes how a part of the UI should look and behave.

UI components such as text, buttons, cards, icons, and input fields are implemented using composable functions like Text(), Button(), Card(), and OutlinedTextField(). These components can be combined together to build complex user interface layouts.

3 Layouts in Compose

Compose provides layout containers that help organize UI components on the screen.

- **Column**: Arranges UI elements vertically.
- **Row**: Arranges UI elements horizontally.

- **Box**: Allows elements to overlap or stack.

These layouts make it easier to structure the UI and control how elements appear on different screen sizes.

4 Codelabs Completed

During the front-end development process, several Compose codelabs were completed to understand how UI components are implemented in Android applications.

These codelabs demonstrated how to:

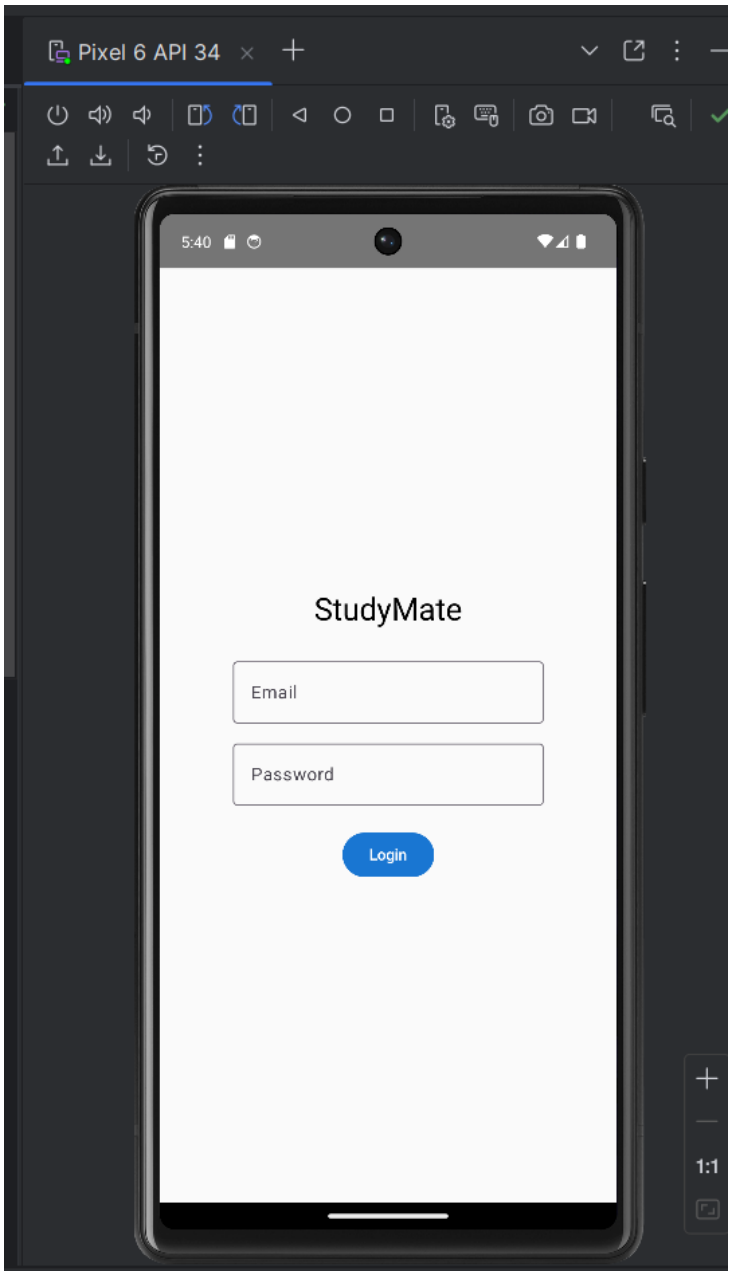
- Create UI components using composable functions
- Use layout containers such as Column and Row
- Manage UI state using remember and mutableStateOf
- Preview UI screens using the @Preview annotation

Completing these codelabs helped build a strong understanding of modern Android front-end development.

5 UI Screens Implemented

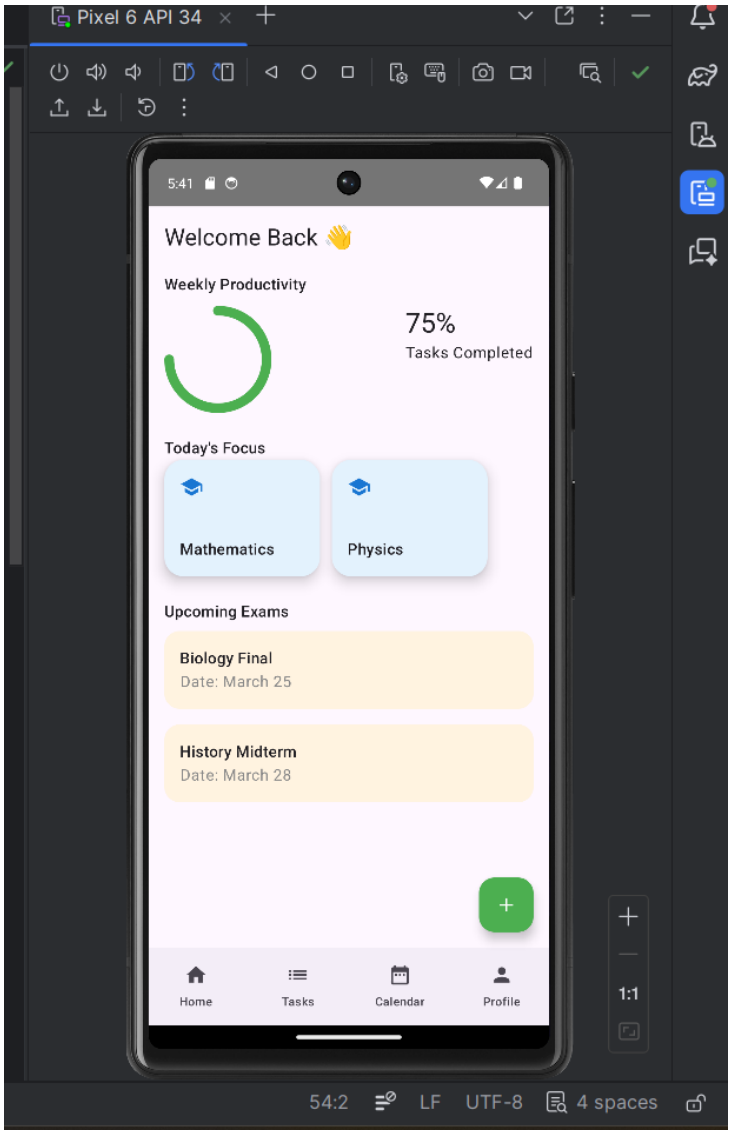
Using the knowledge gained from the codelabs, several UI screens were implemented in Android Studio for the StudyMate application.

5.1 Login Screen



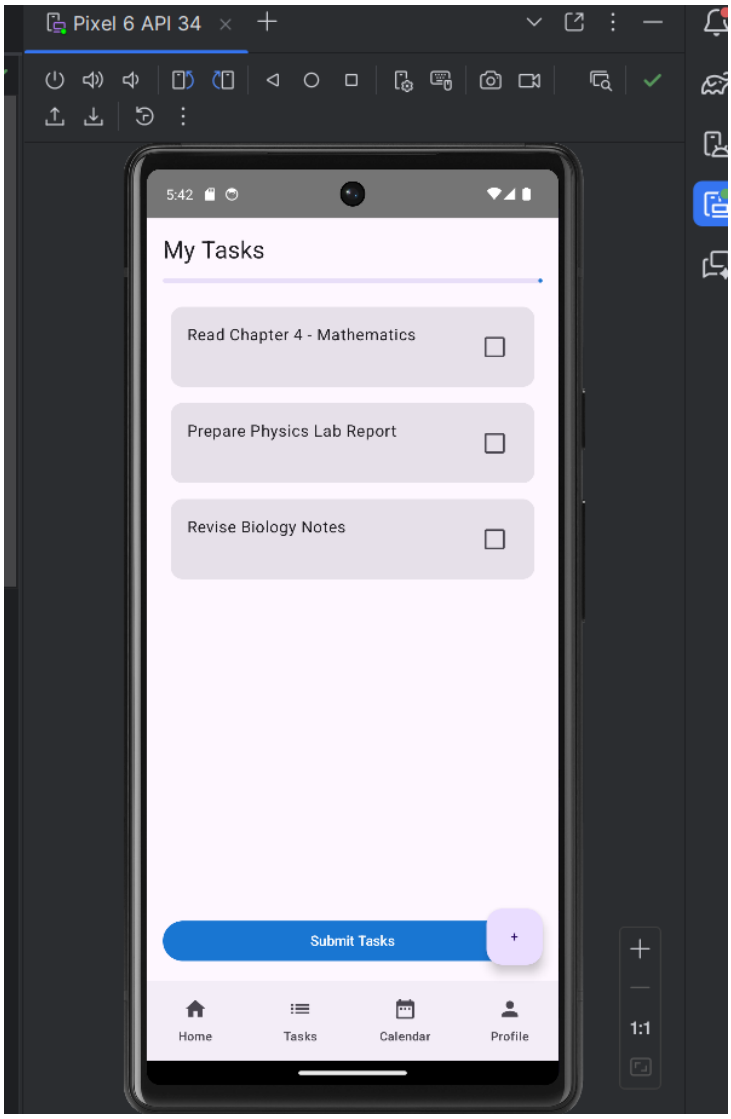
The login screen contains input fields for email and password along with a login button.

5.2 Home Dashboard



The home screen displays study focus cards, productivity indicators, and upcoming exam information.

5.3 Task Management Screen



The task screen allows users to add tasks, complete tasks using checkboxes, and monitor progress.

6 Conclusion

Jetpack Compose simplifies Android UI development by allowing developers to build interfaces using composable functions and declarative programming. The codelabs helped in understanding how layouts, UI components, and state management work together to create responsive user interfaces.

Through this exercise, several UI screens were successfully implemented using Android Studio, demonstrating the practical use of Compose for front-end development.